Grid-Aware Multi-Tenant EV Fleet Charging Orchestration Platform

Full Stack Project — Project Overview Document

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1. Summary

The Grid-Aware EV Fleet Charging Orchestration Platform is a full-stack, multi-tenant web application that lets fleet operators and building managers manage dozens of EV chargers under a shared electrical capacity limit. Built on React, NestJS, PostgreSQL, and real OCPP 1.6-J charger integration, it replaces manual charge scheduling with a live constrained optimizer that balances electrical limits, dynamic electricity pricing, and per-driver priority, updating every dashboard instantly with no page refresh.

2. Problem Statement

Fleet operators and commercial sites running many EV chargers risk tripping the site's transformer/electrical limit — or paying steep demand charges — if every vehicle charges at full power at once. Existing charging apps only handle booking and billing; none solve the live power-allocation problem across shared infrastructure and multiple tenants.

3. Solution

A web platform where each tenant (fleet company) manages its own vehicles and drivers, while a shared optimization engine continuously re-allocates available power across all active charging sessions site-wide — respecting hard electrical limits, live tariff pricing, and per-tenant SLA priority — with a fast fallback so vehicles keep charging even if the optimizer is briefly unavailable.

4. Workflow

Driver Plugs In → OCPP Session Start → Optimizer Allocates Power → Live Dashboard Updates → Session Ends → Tenant Billed

5. User Roles

Site/Fleet Admin	Driver / Tenant Manager
Register chargers and sites Set electrical capacity limits Invite tenant companies View power-allocation reports Configure pricing tiers	View assigned charger and live charge % Set departure time / priority View own fleet's sessions and billing Receive charging-plan-change alerts Download usage invoices

6. Charging Session Board

Every active session is visible on a live board that mirrors the actual charging state across the site:

Queued → Connected → Charging — Optimized → Throttled (Grid Limit) → Complete

7. Features

Authentication	OAuth2/OIDC login, tenant-scoped JWT claims	**Charger Mgmt**	OCPP 1.6-J device registration, remote start/stop
Session	Start, monitor, throttle, and stop charging sessions	**Optimization**	Live MILP solver balancing power cap, tariff, priority, SLA
Billing	Usage-based, tenant-isolated invoicing	**Dashboard**	Site power-usage graph, per-tenant cost, SoC forecast
Notifications	Push/SMS when a charging plan changes	**Multi-Tenancy**	Postgres Row-Level Security, tenant-scoped WebSocket channels

8. Real-Time Sync (WebSockets + OCPP)

Charger firmware speaks the real OCPP 1.6-J protocol to the gateway service. Every meter reading and status change is broadcast over tenant-scoped WebSocket channels, so a driver's app and the fleet dashboard reflect the charger's actual state within seconds — no polling.

Charger (OCPP) → OCPP Gateway → Session Service → Optimizer (60s cycle) → WebSocket Broadcast → Dashboard / Driver App

9. Database Design

Table	Fields
Tenants	Tenant ID, Company Name, Site ID, Billing Plan
Chargers	Charger ID, Site ID, OCPP Endpoint, Max Power, Status
Sessions	Session ID, Charger ID, Vehicle ID, Tenant ID, Start/End, kWh, Allocated Power
Vehicles	Vehicle ID, Tenant ID, Battery Capacity, Driver, Priority Tier
Invoices	Invoice ID, Tenant ID, Period, Total kWh, Amount

10. Tech Stack

Frontend	React, Next.js (driver app), Tailwind CSS	**Backend**	NestJS (or FastAPI), OCPP.js gateway
Database	PostgreSQL (Row-Level Security)	**Real-time**	WebSockets, Redis Pub/Sub
Optimization	MILP via PuLP / Google OR-Tools	**Auth**	OAuth2 / OIDC, tenant-scoped JWT
Billing	Metered usage engine (Stripe-style)	**Deployment**	Docker, AWS ECS/EKS, API Gateway
Monitoring	Prometheus, Grafana	**Testing**	Charger simulator for load/fault testing

11. Future Scope

- Vehicle-to-grid (V2G) discharge scheduling during peak demand
- Solar/battery microgrid integration at the site level
- Predictive charging-demand forecasting per tenant
- Dynamic carbon-intensity-aware charging windows
- Marketplace for tenants to trade unused power allocation